

Name: _____

Please **CIRCLE** the tutorial session you attend.

Mon. 1.30-2.20; 2.30-3.20; 3.30-4.20; 4.30-5.20; 5.30-6.20

Tue. 9.30-10.20; 10.30-11.20; 11.30-12.2

Each question is worth 1 mark

1. Which of the following is an example of panel data?
 - a. Data on the monthly interest rates in Singapore from January 2020 to March 2024
 - b. Data on the yearly consumption of bubble tea by 1000 households in the year 2023.
 - c. Data on the number of vacancies in various government agencies on a particular month.
 - d. Data on the unemployment rates in different cities of China from January 2020 to March 2024.
2. Covariance stationary sequences where $Cov(y_t, y_{t+h}) \rightarrow 0$ as $h \rightarrow \infty$ are said to be:
 - a. AR processes
 - b. dynamic processes
 - c. weakly dependent
 - d. Independent
3. Which of the following models is appropriate when the dependent variable is binary?
 - a. Probit model
 - b. Logit model
 - c. LPM
 - d. All of the above
4. In the Probit model, the G function is the:
 - a. normal probability distribution function.
 - b. standard normal cumulative distribution function.
 - c. standard normal probability distribution function.
 - d. cumulative distribution of normal distribution function.
5. Which of the following statements is true regarding the marginal effects calculated in the Probit model and the Logit model?
 - a. Both models do not allow for calculation of partial effects.
 - b. Partial effects in the probit model are easier to interpret than in the logit model.
 - c. Partial effects in the logit model are easier to interpret than in the probit model.
 - d. Partial effects in both models are difficult to interpret because they depend on the value of the independent variable.
6. Consider an estimated probit model: $\hat{y} = \Phi(\hat{\beta}_0 + \hat{\beta}_1 x) = \Phi(0 + 2x)$. The partial effect $\frac{d\hat{y}}{dx}$ evaluated at $x = 0$ is roughly equal to
 - a. 0.4
 - b. 0.8
 - c. 1.0
 - d. None of the above
7. Consider a regression model: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. Suppose z_1 and z_2 are valid instruments for x_1 and x_2 , respectively. Which of the following conditions is correct?
 - a. $Cov(z_1, x_2) \neq 0$
 - b. $Cov(z_2, x_1) \neq 0$
 - c. $Cov(z_1, z_2) \neq 0$
 - d. $Cov(z_1, u) = 0$
8. Consider a regression model: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. Suppose z is an instrument for x_1 . Which of the following conditions denotes instrument relevance?
 - a. $Cov(z, x_1) \neq 0$
 - b. $Cov(z, x_2) \neq 0$
 - c. $Cov(x_1, u) \neq 0$
 - d. $Cov(x_1, x_2) \neq 0$

9. Consider a regression model: $y = \beta_0 + \beta_1 x + u$, and z is IV for the endogenous x . The variance of the IV estimator is larger than that of the OLS estimators because ____.
- a. $\rho_{z,x}^2 > 1$ b. $\rho_{z,x}^2 = 1$ c. $\rho_{z,x}^2 < 1$ d. $\rho_{z,x}^2 = 0$
10. Consider a simple regression model: $y = \beta_0 + \beta_1 x + u$, where z is an instrument for x . If $Cov(z, u) = 0$ and $Cov(z, x) \neq 0$, the value of β_1 in terms of population covariances is ____.
- a. $\frac{Cov(z,y)}{Cov(z,x)}$ b. $\frac{Cov(z,u)}{Cov(z,x)}$ c. $\frac{Cov(z,x)}{Cov(z,y)}$ d. $\frac{Cov(z,y)}{Var(x)}$

True/False Questions

- ____ 11. The linear probability model assumes homoskedasticity, where the variance of the error term is constant across all observations.
- ____ 12. If the time series $\{y_1, y_2, \dots, y_T\}$ is white noise, then the y observations are serially uncorrelated.
- ____ 13. Suppose there are two instruments z_1 and z_2 for two endogenous variables x_1 and x_2 , then we can use the Sargan test to test for instrument exogeneity.
- ____ 14. The 2SLS estimator is always more efficient than the OLS estimator.
- ____ 15. A strong instrument variable is highly correlated with the endogenous explanatory variable but not necessarily with the dependent variable.

Bonus Questions

16. $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ and $\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ are **both** consistent estimators for the population covariance X and Y , i.e., $Cov(x, y)$.
- a. True
b. False
c. Not enough information
17. Which of the following is a **true** description regarding the MAS's article "Returns to Education in Singapore"?
- a. The authors used the IQ score as a proxy for ability to solve the endogeneity problem.
b. The authors used the number of siblings as an instrument to solve the endogeneity problem.
c. The authors used the panel data models to solve the endogeneity problem.
d. The authors used robust standard errors in their regression analysis.
e. None of the above